

Deva Anand

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Education

University of Massachusetts Amherst

Master of Science in Computer Science

September 2025 – May 2027 (Expected)

Amherst, MA

Coursework: Advanced Machine Learning, Advanced NLP, Performance Engineering, Systems for Data Science, Optimization in CS

Vels Institute of Science, Technology and Advanced Studies

Bachelor of Engineering in Computer Science and Engineering

August 2019 – May 2023

Chennai, India

Skills

Languages

Python, TypeScript, JavaScript, C++, Java, SQL, Bash

App & Frontend

React, Flutter, Electron, REST API integration, async programming, caching, performance profiling

ML & Model Families

CNN, Transformer, diffusion, LLMs, multimodal (vision / speech), PyTorch, HuggingFace Transformers, NumPy

LLM & Integration

LLM agents, tool use, provider routing/fallback, prompt design, OpenAI / Groq APIs, RAG

Backend & Cloud

FastAPI, Node.js, Express, REST APIs, Git, GitHub Actions, Docker, AWS, GCP, Azure

Experience

Wysa

Backend Engineer

July 2023 – July 2025

Bengaluru, India

- Architected a full-stack app (**React, Node.js, MongoDB**) used daily by the AI team to label data for production models, replacing spreadsheet workflows with secure role-based REST APIs; cut manual work by **100+ hours/month** and improved throughput by **60%**.
- Engineered a multi-tenant NHS eTriage backend (Node.js, MongoDB) for **8+ UK clients** processing **10,000+ monthly submissions**; integrated key user flows with external APIs and owned reliability, monitoring production logs and shipping fixes for tenant-specific incidents.
- Took part in a measured **GitHub Copilot** productivity pilot, routing tech-debt PRs, code migrations, and security-finding triage through AI-assisted workflows; maintained a versioned `copilot-instructions.md` conventions file to keep AI-generated code aligned with codebase patterns.

Cario Growth Services Pvt. Ltd.

Backend Developer Intern

November 2022 – May 2023

Chennai, India

- Integrated **LLMs** into a chat product to generate context-aware replies from conversation history and user input, iterating on prompt design and response-quality heuristics, while building **20+ Node.js + Fastify REST APIs** over PostgreSQL.
- Shipped an open-source **LLM auto-commenting feature in production (4-bit quantized Llama-7B via llama.cpp)** on a single VM) with per-character persona prompts, async queued generation, rate limits, and a content-safety filter, built end-to-end with no ML mentor on the team.

Projects

Sonare - Cross-Platform App with On-Device AI

github.com/Deva-1903/Qualcomm-Sep25-Team-Sonare

React, Electron, FastAPI, MediaPipe, whisper.cpp | Qualcomm Edge AI Hackathon 2025

- Built a **cross-platform app** integrating camera, microphone, UI, and backend into an **on-device pipeline** (MediaPipe hand tracking, gesture classification, whisper.cpp ASR, local TTS), a multimodal vision/speech experience prototyped and shipped end-to-end under hackathon pressure.
- Engineered a FastAPI backend and **Electron/React frontend** running fully offline with no network dependencies; designed **async APIs and queues** to keep the UI responsive while handling local video and audio streams under on-device inference constraints.

AgenticSearch - AI Agent with LLM Integration & Verification

github.com/Deva-1903/ciir_agentic_search

Python, FastAPI, Brave Search, OpenAI / Groq, SQLite

- Built a multi-stage **AI agent** (Python) that integrates LLMs with deterministic parsers: plans typed retrieval, reranks pages, extracts entities with LLM fallback, and verifies AI-generated output, returning cell-level provenance for every value.
- Hardened the service with **150+ automated tests**, provider routing/fallback across LLM backends, and pipeline instrumentation, reasoning about model limitations to keep extraction predictable.

Underdogs Fitness - 0→1 Production App (Live 3+ Years)

underdogsfitness.in

MongoDB Atlas, Express, React + Redux, Node.js, Stripe

- Shipped a **0→1 production app** solo end-to-end (React + Redux frontend, Node.js/Express backend): auth, membership and attendance, **Stripe payments with a partial-payment ledger**, and cron automation, **live and actively used 3+ years**.
- Designed it multi-branch from day 1 with clean, modular data scoping; when the client expanded a year later it onboarded as a config change with **zero schema migration**, deployed via git-webhook automation.

Publication & Extracurriculars

- Publication:** “Alzheimer’s Disease Classification using Transfer Learning,” **IEEE CONIT 2023**, applied deep transfer learning on neuroimaging data for medical image classification.
- Selected for **GirlScript Summer of Code (GSSoC) 2023**; contributed to multiple open-source projects (Linkfree, Freehit, ProjectsHut).
- Tutored **30+** underprivileged students in programming at Sayur, non-profit in South Tamil Nadu.